

1.

- a. Convolutional Layer => Extract features from input
 Pooling Layer => Downsample feature maps for efficiency
 FC Layer => Map extracted features to final predictions
 Activation Layer => Increase non-linearity of the network
 Loss Function => Evaluate error for classification or regression
- b. B
- c. B
- d. B

2.

a.

Concept	PF	KF
Represents belief with multiple samples (particles)		
Assumes Gaussian noise and linear system		
Uses motion models and measurement models		
Handles non-linear and non-Gaussian models naturally		
Uses a single mean and covariance matrix		
Performs prediction-update (Bayes filtering cycle)		
May require resampling to focus on high-probability particles		
Uses an optimal gain to balance prediction and measurement		
Requires an initial estimate (initial belief)		
Can be used for robot localization and SLAM		

- b. B
- c. Because robots operate in noisy and unknown environments and you need probability to make smart decisions to account for motion and sensor inaccuracies.

3.

- a. C
- b. B
- c. B

d. $G_t = \frac{dg}{d(x, y, \theta)} = \left[\frac{d}{dx}, \frac{d}{dy}, \frac{d}{d\theta} \right] =$

$$\begin{bmatrix} 1 & 0 & -d\sin\theta \\ 0 & 1 & d\cos\theta \\ 0 & 0 & 1 \end{bmatrix}$$

e. Computed is:

1	0	-0.5
0	1	0.866
0	0	1

4.

- a. B
- b. B
- c. B
- d. B
- e. B
- f. B

5.

- a. B
- b. C
- c. B
- d. A
- e. The Bellman Equation is the value of a state as the sum of the reward from taking the best possible action and the discounted value of the next state.

$$V(s) = \max(a) \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V(s')]$$

- f. Policy iteration alters between improvement and policy evaluation while value iteration updates estimates for value and then derives the policy